

SECTION A: SET THEORY, REAL NUMBER AND MATHEMATICAL INDUCTION

1. _____ is used to prove the validity of proposition of the set of non- negative integers.

A. Real number system

B. Real line

C. Associativity

D. The principle of mathematical induction+++

2. Which of the following statements is false for all positive integers using the principle of mathematical induction.

A. $2^n > n$

B. $5^{2n} + 3n + 1$ is a multiple of 9

C. $5^n - 1$ is a multiple of 4

D. $7^{2n+1} + 1$ is a multiple of 3+++

3. A student of the Department of Mathematical Sciences, Federal University of Technology, Akure was asked to show that $5^n - 1$ is divisible by 4 for all positive integers k using the principle of mathematical induction. Which of the following mathematical statements is correct for all positive integers M?

A. $f(1) = 8$

B. $f(M) = 4(5M + 1)$

C. $f(k + 1) = 4(5M + 1)+++$

D. $f(k) = 4(5M + 1)$

4. Which of the following is not one of the properties of real numbers?

A. existence of the multiplicative inverse

B. closure

C. it satisfies the principle of mathematical induction+++

D. existence of the zero of the set

5. Let a, b, c and d be real numbers. Which of the following statements is false?

A. if $a > b$, then $a + c < b + c$ +++

B. if $a > b$ and $c > 0$, then $ac > bc$

C. if $a > b$ and $c < 0$, then $ac < bc$

D. if $a > b$ and $c > d$, then $a + c > b + d$

6. A 100L student of the department of Biochemistry, Federal University of Technology, Akure, Samuel Victory, was asked to use the principle of mathematical induction to prove that $\sum_{i=0}^{i=n} x^i = \frac{1-x^{n+1}}{1-x}$. For what values of x can she prove this?

A. for all real numbers x

B. for all real numbers x satisfying $x \neq 1$ +++

C. for all real numbers x satisfying $x = 1$

D. for all positive integers n

7. Mr. Dansu's project student was asked to prove that $f(n) = n^4 + 4n^2 + 11$ is divisible by 16 for all odd positive integers. What are the possible values of n?

A. 1, 2, 3, ..., k, k+1 where k is a positive integer

B. 1, 2, 3, ..., k, k+1, ..., ∞ where k is a positive integer

C. 1, 3, ..., 2k-1, 2k+1, ... where k is a positive integer +++

D. none of the above

8. In proving that $f(n) = 7^{2n+1} + 1$ is a multiple of 8, Mr. Olodo assumed that the proposition is valid when $n=k$ where k is a positive integer. Which of the following statements is correct?

A. $f(K + 1) = 344$

B. $f(K + 1) = 8(49M - 6)$ where M is a positive integer +++

C. $f(K + 1) = 8M - 1$
where M is a positive integer

D. $f(K + 1) = 8(49M + 6)$ where M is a positive integer

9. Which of the following laws of algebra of addition and multiplication of natural numbers is incorrect for all $x, y, z \in \mathbb{N}$

A. Closure: (i) $x + y \in \mathbb{N}$ (ii) $x \cdot y \in \mathbb{N}$

B. Commutative: (i) $x + y = y + x$
(ii) $x \cdot y = y \cdot x$

C. Associative: (i) $x + (y + z) = (x + y) + z$
(ii) $x \cdot (y + z) = (x \cdot y) \cdot z$ +++

D. Distributive: (i) $x \cdot (y + z) = x \cdot y + x \cdot z$
(ii) $(y + z) \cdot x = y \cdot x + z \cdot x$

10.

Given that for all positive integer values of n , $5^{2n} + 3n - 1$ is an integer multiple of 9. If k is a positive integer, then by the principle of mathematical induction

A. $f(k + 1) = 9(25M + 8k + 3)$
where M is a positive integer

B. $f(k + 1) = 9(25M - 8k + 3)$
where M is a positive integer +++

C. $f(k + 1) = 5^{2(k+1)} + 3k + 2$
where M is a positive integer

D. $f(k) = 5^{2(k+1)} + 3k + 2$ where M is a positive integer

11. Which of the following is the sequential arrangement of the application of the principle of mathematical induction? (i) If T_1 is true (ii) then T_n is true for every positive integer n (iii) for every positive integer k , the truth of T_k would imply the truth of T_{k+1} ,.

- A. ii, i and iii
- B. i, ii and iii
- C. i, iii and ii +++
- D. iii, ii and i

12. Given that $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$. If k is a positive integer, then by the use of the principle of mathematical induction,

- A. $f(k+1) = \frac{k+3}{(k+1)(k+2)}$ +++
- B. $f(k+1) = \frac{k+3}{(k+1)(k+3)}$
- C. $f(k+1) = \frac{1}{(k+1)(k+2)}$
- D. $f(k+1) = \frac{k}{k+1}$

13. Given that $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$. If k is a positive integer, then by the use of the principle of mathematical induction,

- A. $f(k+1) = \frac{k+1}{2k+3}$ +++
- B. $f(k+1) = \frac{k}{2k+3}$
- C. $f(k+1) = \frac{k}{2k+1}$
- D. $f(k+1) = \frac{1}{(2k+1)(2k+3)}$

14. Given that $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \dots + \frac{1}{x^n} = \frac{1}{x-1} - \frac{1}{x^n(x-1)}$. If k is a positive integer, then by the use of the principle of mathematical induction,

- A. $f(k+1) = \frac{1}{x-1} - \frac{1}{x^{k+1}(x-1)}$
- B. $f(k+1) = \frac{1}{x-1} - \frac{1}{x^k(x-1)}$
- C. $f(k+1) = \frac{x^k-1}{x^{k+1}(x-1)}$
- D. $f(k+1) = \frac{x^{k+1}-1}{x^{k+1}(x-1)}$ +++

15. Given the set $A = \{a, b, c, d\}$; which of the following is not an element of the power set of A .

- A. $\{a, b\}$
- B. $\{\emptyset\}$ +++
- C. A
- D. $\{a, b, c\}$

16. How many subsets will a set containing 5 elements have?

- A. 32
- B. 25
- C. 36
- D. 64 +++

17. If $\mu = \{e, f, g, h, p, q, r, s\}$ be the universal set, $M = \{e, h, q, s\}$ and $N = \{h, p, r\}$. Find $M^c \cup N^c$.

- A. $\{f, p, r\}$
- B. $\{e, f, q, s\}$
- C. $\{e, f, g, p, q, r, s\}$ +++
- D. $\{e, g, p, q, r, s\}$

18. P and Q are subsets of the universal set μ defined as $\mu = \{x: x \text{ is an integer and } 1 < x < 15\}$

$P = \{x: x \text{ is odd}\}$ and $Q = \{x: x \text{ is prime}\}$. Find the cardinal number of the intersection

of the compliments of P and Q .

A. 3

B. 4

C. 5

D. 6+++

E. 9

19. If $P = \{1,2,3,4,5,6,7,8\}$, $Q = \{1,4,9\}$, $R = \{2,4,8\}$. Find $(P \cap Q) \cup R$.

A. $\{1,2,4,8\}$

B. $\{1,2,4,8,9\}+++$

C. $\{1,2,4,7,8\}$

D. $\{1,2,3,4,5,6,7,8\}$

20. In a class of 40 students, 24 play football, 18 play volleyball and 6 do not play any game. How many Students play both Football and Volleyball?

A. 2

B. 7

C. 8+++

D. 9

21. Four members of a school first eleven cricket team are also members of the first fourteen

Rugby team. How many boys play in only one of the two teams?

A. 25

B. 21

C. 17+++

D. 29

22. If $S = \{x: x^2 = 9, x < 2\}$ then S is equal to

A. $\{-3\}+++$

B. $\{3\}$

C. $\emptyset$

D. $\{-3,3\}$

23. A School of 38 teachers has 15 married women. If 6 of the teachers are couples, how many of the teachers are neither married nor have their spouses not in the school?

A. 17

B. 23

C. 20+++

D. 21

24. Which of the following is equivalent to $[P^c \cup (Q \cap Q^c)]$

A. P

B. P^c+++

C. Q

D. Q^c

25. Given that the sets A and B are partitioned sets of E; then $A \cap (E \cap B)^c$ is

A. $A+++$

B. $\emptyset$

C. B

D. E

26. Which of the following sets is equivalent to $(P \cup Q) \cap (P \cap Q^c)$

A. $P \cup Q$

B. $P \cup Q^c$

C. $P \cap Q^c+++$

D. $\emptyset$

27. P and Q are subsets of the universal set U defined as $U = \{x: x^3 - x^2 - x + 1 = 0\}$

$P = \{x: x \text{ is an integer}\}$ and $Q = \{x: x \text{ odd}\}$, find $P \cap Q$.

A. $\{-1,1,1\}$

B. $\{-1,1\}+++$

C. $\{1,1,-1\}$

D. $\{-1,-1\}$

28. If $E = \{1,2,3,4\}$ and $A = \{1,3,5\}$, the symmetric difference of E and A written $A \Delta E$ is

A. $\{1,2\}$

B. $\{2,4,5\}+++$

C. $\{1,2,3,4,5\}$

D. $\{1,3\}$

29. If the universal set $U = \{x: x \text{ is a natural number and } 1 \leq x \leq 9\}$, $P = \{x: 1 \leq x < 4\}$ and

$Q = \{2,4,6,8\}$. Find $(P \cup Q)^c$

A. $\{1,2,3,4,5,6,7,8,9\}$

B. $\{1,2,3,4,6,8\}$

C. $\{1,5,6,7\}$

D. $\{5,7,9\}+++$

30. In a science class of 41 students, each student offers at least one of Mathematics and

Physics. If 22 Students offer Physics and 28 Students offer Mathematics, how many

Students offer Physics only

A. 19

B. 9

C. 13+++

D. 14

31. If $U = \{0, 2,3,6,7,8,9,10\}$ is the universal set , $E = \{0,6,8,10\}$ and $F = \{x: x^4=14\}$. Find

$(E \cup F)^c$

A. $\{3,7,9\}+++$

B. $\{2,3,7,9\}$

C. $\{3,7,9\}$

D. $\emptyset$

32. If $X = \{\text{all prime factors of } 44\}$ and $Y = \{\text{all prime factors of } 60\}$.The elements of $X \cup Y$

and $X \cap Y$ respectively are :

A. $\{2,4,3,5,11\}$ and $\{4\}$

B. $\{4,3,5,1\}$ and $\{3,4\}$

C. $\{2,5,11\}$ and $\{2\}$

D. $\{2,3,5,11\}$ and $\{2\}+++$

33. Simplify $(A \cup B)^c \cap (A \cap B^c)$

A. $(A^c \cup B)$

B. $A \cup B^c$

C. $(A \cup B)^c$

D. $\phi+++$

34. Simplify $(A \cup B)^c \cap A^c$

A. $(A \cup B)$

B. $A \cap B^c$

C. $A^c \cap B$

D. $(A^c \cap B^c)+++$

35. In a village all the people speak Hausa or English or both , if 56% speak Hausa and 63% speak English ,what percentage speak both languages

A. 19%+++

B. 119%

C. 62%

D. 38%

36. is a set that contains another set.

- A. Power set
- B. Super set+++
- C. Subset
- D. Proper Set

37. In a youth club with 94 members, 60 like modern music and 50 like traditional music. The number of members who like both traditional and modern music is thrice those who do not like any type of music. How many members like only one type of music?

- A. 8
- B. 24
- C. 62+++
- D. 86

38. The symmetric difference of A and B expressed $A \Delta B$ is equal to

- A. $(A-B) \cap (B-A)$
- B. $(A-B) \cup (B-A)+++$
- C. $(A \cap B) \cup (B \cap A)$
- D. $(A \cup B) \cap (B \cup A)$

39. $\xi = \{e, f, g, h, p, q, r, s\}$, $M = \{e, h, q, s\}$ and $N = \{h, p, r\}$. Find $M^c \cup N^c$

- A. $\{f, p, r\}$
- B. $\{e, f, q, s\}$
- C. $\{e, f, g, p, q, r, s\}+++$
- D. $\{e, g, p, q, r, s\}$

40. How many subsets will a set containing 6 elements have?

- A. 25
- B. 32
- C. 36
- D. 64+++

41. Which of the following is equivalent to $\{P^c \cap (Q \cup Q^c)\}$

- A. P
- B. P^c+++
- C. Q
- D. ϕ

42. Which of the following options below is true

- A. $A \cup A^c = A$
- B. $A \cap A^c = A$
- C. $(A^c)^c = U$
- D. $nP(A) = 2^{n(A)}+++$

43. Out of 25 teachers, 16 are married and 15 are women. If 6 of the men are married, how many of the women are not married?

- A. 6
- B. 10
- C. 5+++
- D. 9

44. Which of the following is not a true property of a Cartesian product?

- A. $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$
- B. $(A \cup B) \times (C \cup D) = (A \times C) \cup (B \times D)$
- C. $(A \cup B) \times C = (A \times C) \cup (B \times C)$
- D. $A \times \phi = \phi \times A = A+++$

45. If $(A \cap B^c) \cup (A^c \cap B) = A \Delta B$, what does the symbol " Δ " signifies?

- A. "difference"
- B. "asymmetric difference"
- C. "symmetric difference"+++
- D. "union"

46. Let $U = \{1, 2, 3, 4, 5, 6\}$ be the universal, $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$. Find $A^c \cup (B \cup B^c)$.

- A. $\emptyset$
- B. $\{1, 2, 3\}$
- C. $\{4, 5, 6\}$
- D. U +++

47. $C = \{1, 2, 3, 4, \dots\}$. What is the name of the set C ?

- A. Finite set
- B. Closed set
- C. Universal set
- D. Infinite set+++

48. What does the set $\{x: x \notin A \text{ and } x \in B\}$ defines?

- A. set containing elements in A and not in B
- B. set containing elements not in A but in B +++
- C. set containing elements both in A and B
- D. set containing elements both in A and B

49. When a set $C = \{ \}$ it means that C is_____

- A. a universal set

B. C is finite set

C. C is an empty set+++

D. is a universal set

50. $A \subseteq B$ and $B \subseteq C$ implies $A \subseteq C$ is the

A. Complementary law

B. transitivity Law+++

C. Inverse law

D. commutativity law

SECTION B: CIRCULAR MEASURE AND TRIGONOMETRIC FUNCTIONS

1. Evaluate $\frac{\sin^2 \theta}{\cos^2 \theta - 1}$.
 A. -1
 B. 1
 C. 2
 D. -2
2. Simplify $\frac{\sin^2 x}{\tan x}$.
 A. $\sin x$
 B. $\cos x$
 C. $\sin x \cos x$
 D. $\sin^2 x \cos^2 x$
3. If $\tan \frac{1}{2}x = b$, express $\tan x$ in terms of b .
 A. $2b$
 B. $1 - b^2$
 C. $\frac{1-b^2}{2b}$
 D. $\frac{2b}{1-b^2}$
4. If $\tan \frac{1}{2}x = f$, what is $\sin x$ in terms of f ?
 A. $\frac{2f}{1+f^2}$
 B. $\frac{2f}{1+f}$
 C. $\frac{2f^2}{1+f}$
 D. $\frac{2f^2}{1+f^2}$
5. If $\tan \frac{1}{2}x = k$, what is $\cos x$ in terms of k ?
 A. $\frac{2k^2}{1+k^2}$
 B. $\frac{1-2k^2}{1+k^2}$
 C. $\frac{1-k^2}{1+k^2}$
 D. $\frac{1+k^2}{1-k^2}$
6. What is the result of $1 + \tan^2 \theta$?
 A. $\sin^2 \theta$
 B. $\sec^2 \theta$
 C. $\cos^2 \theta$
 D. $\tan^2 \theta$
7. Evaluate $1 + \cot^2 \theta$.
 A. $\sin^2 \theta$
 B. $\operatorname{cosec}^2 \theta$
 C. $\cos^2 \theta$
 D. $\tan^2 \theta$
8. Simplify $\sin^2 p + (1 + \cos^2 p)^2$.
 A. $2(1 + \cos p)$
 B. $2 \cos p$
 C. $1 + \cos p$
 D. $\cos p - 1$
9. Evaluate $\frac{1+\sin \emptyset}{1+\cos \emptyset} \times \frac{1+\sec \emptyset}{1+\operatorname{cosec} \emptyset}$.
 A. $\sin \emptyset$
 B. $\cos \emptyset$
 C. $\tan \emptyset$
 D. $\cot \emptyset$
10. Expand $\tan(45^\circ + A)$.
 A. $\frac{1+\tan A}{1-\tan A}$
 B. $\frac{1-\tan A}{1+\tan A}$
 C. $\frac{\tan A}{1+\tan A}$
 D. $\frac{1+\tan A}{1-\tan A}$
11. Evaluate $\tan(90^\circ + A)$.
 A. 0
 B. ∞
 C. -1
 D. 1
12. If $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$; $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$, evaluate $\sin 75^\circ$.
 A. $\frac{\sqrt{2}+\sqrt{6}}{4}$
 B. $\frac{\sqrt{2}-\sqrt{6}}{4}$
 C. $\frac{\sqrt{6}-\sqrt{2}}{4}$

D. $\frac{\sqrt{2}+\sqrt{3}}{4}$

13. If $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$; $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$, evaluate $\cos 75^\circ$. A.

$\frac{\sqrt{6}+\sqrt{2}}{4}$

B. $\frac{\sqrt{2}+\sqrt{3}}{4}$

C. $\frac{\sqrt{6}-\sqrt{2}}{4}+++$

D. $\frac{\sqrt{2}-\sqrt{6}}{4}$

14. Find $\cos 105^\circ$, if $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$; $\sin 60^\circ = \frac{\sqrt{3}}{2}$; $\cos 60^\circ = \frac{1}{2}$

A. $\frac{\sqrt{2}+\sqrt{6}}{4}$

B. $\frac{\sqrt{2}-\sqrt{6}}{4}+++$

C. $\frac{\sqrt{2}-\sqrt{6}}{5}$

D. $\frac{\sqrt{2}-\sqrt{3}}{4}$

15. Which of these evaluates to 1?

A. $\sec^2 x - \tan^2 x+++$

B. $\sec^2 x + \tan^2 x$

C. $\tan^2 x - \sec^2 x$

D. $\tan x - \sec^2 x$

16. Which of these evaluates to 1?

A. $\cot y - \operatorname{cosec} y$

B. $\cot^2 y - \operatorname{cosec}^2 y$

C. $\operatorname{cosec} y - \cot y$

D. $\operatorname{cosec}^2 y - \cot^2 y+++$

17. Evaluate $\sec^2 p - 1$.

A. $\tan p$

B. $\tan p \sec p$

C. $\tan^2 p+++$

D. $\tan^3 p$

18. Evaluate $\tan^2 q - \sec^2 q$.

A. 1

B. $-1+++$

C. 2

D. -2

19. What is the result of $\sec^2 x - \tan^2 x$?

A. $1+++$

B. -1

C. 2

D. -2

20. Which of these evaluates to -1 ?

A. $\cot^2 r + \operatorname{cosec}^2 r$

B. $\cot r$

C. $\cot^2 r - \operatorname{cosec}^2 r+++$

D. $\cot r - \operatorname{cosec} r$

21. Which of these is equivalent to $\sin 2w$?

A. $2 \sin w \cos w+++$

B. $2 \sin^2 w$

C. $2 \cos^2 w$

D. $\sin w \cos w$

22. Which of these is equivalent to $\cos 2z$?

A. $1 - \cos^2 z$

B. $1 + \cos z$

C. $2 \cos^2 z - 1+++$

D. $1 + 2 \cos^2 z$

23. Which of these is equivalent to $\tan 2t$?

A. $\frac{1-\tan^2 t}{t}$

B. $\frac{2 \tan t}{1-\tan^2 t}+++$

C. $\frac{1-\tan^2 t}{2 \tan t}$

D. $\frac{\tan t+1}{2 \tan t}$

24. Expand $\sin v$.

A. $\sin \frac{1}{2} v \cos \frac{1}{2} v$

B. $\sin^2 \frac{1}{2} v$

C. $\cos^2 \frac{1}{2} v$

D. $2 \sin \frac{1}{2} v \cos \frac{1}{2} v+++$

25. Expand $\cos a$.

- A. $\cos^2 \frac{a}{2} + 1$
- B. $2\cos^2 \frac{a}{2} - 1$
- C. $2\cos^2 \frac{a}{2} + 1$
- D. $2 - \cos^2 \frac{a}{2}$

26. What is the equivalence of $\sin(p + q)$?

- A. $\sin p \sin q + \cos p \cos q$
- B. $\sin p \sin q - \cos p \cos q$
- C. $\sin p \cos q - \cos p \sin q$
- D. $\sin p \cos q + \cos p \sin q$

27. What is the equivalence of $\sin(p - q)$?

- A. $\sin p \sin q + \cos p \cos q$
- B. $\sin p \sin q - \cos p \cos q$
- C. $\sin p \cos q - \cos p \sin q$
- D. $\sin p \cos q + \cos p \sin q$

28. Expand $\cos(x + y)$.

- A. $\cos x \cos y - \sin x \sin y$
- B. $\cos x \cos y + \sin x \sin y$
- C. $\sin x \sin y - \cos x \cos y$
- D. $\sin x \sin y + \cos x \cos y$

29. Expand $\cos(x - y)$.

- A. $\cos x \cos y - \sin x \sin y$
- B. $\cos x \cos y + \sin x \sin y$
- C. $\sin x \sin y - \cos x \cos y$
- D. $\sin x \sin y + \cos x \cos y$

30. Express $\sin 60^\circ$ in surd form.

- A. $\frac{1}{2}$
- B. $\frac{\sqrt{3}}{2}$
- C. $\frac{1}{\sqrt{2}}$
- D. 0

31. Express $\sin 45^\circ$ in surd form.

- A. $\frac{1}{2}$
- B. $\frac{\sqrt{3}}{2}$
- C. $\frac{1}{\sqrt{2}}$

D. 0

32. Which of these is equal to $\sin \theta$?

- A. $1 - \cos \theta$
- B. $\cos(90^\circ - \theta)$
- C. $\sin(90^\circ - \theta)$
- D. $\cos \theta - 1$

33. Which of these is equivalent to $\cos 13^\circ$?

- A. $\sin 77^\circ$
- B. $\cos 77^\circ$
- C. $\sin 87^\circ$
- D. $\cos 87^\circ$

34. Which of these is equivalent to $\cos \beta$?

- A. $\cos(90^\circ - \beta)$
- B. $\sin(90^\circ - \beta)$
- C. $\sec(90^\circ - \beta)$
- D. $\operatorname{cosec}(90^\circ - \beta)$

35. $\sin 81^\circ$ is the same as

- A. $\cos 9^\circ$
- B. $\sin 9^\circ$
- C. $\tan 9^\circ$
- D. $\sec 9^\circ$

36. What is the equivalence of $\tan(A + B)$?

- A. $\frac{\cos(A+B)}{\sin(A+B)}$
- B. $\frac{1}{\tan(A+B)}$
- C. $\frac{\tan A + \tan B}{1 - \tan A \tan B}$
- D. $\frac{\tan A - \tan B}{1 + \tan A \tan B}$

37. Which of the following is the same as $\tan(A - B)$?

- A. $\frac{\cos(A+B)}{\sin(A+B)}$
- B. $\frac{1}{\tan(A+B)}$
- C. $\frac{\tan A + \tan B}{1 - \tan A \tan B}$
- D. $\frac{\tan A - \tan B}{1 + \tan A \tan B}$

38. What is the equivalence of $\tan 90^\circ$?

A. 0

B. 1

C. -1

D. ∞

39. If $\cos \delta = \frac{a}{b}$, find $1 + \tan^2 \delta$.

A. $\frac{b}{a}$

B. $\frac{b^2}{a}$

C. $\frac{a^2}{b}$

D. $\frac{b^2}{a^2}$

40. If $\cos \delta = \frac{a}{b}$, find $1 - \tan^2 \delta$.

A. $\frac{b}{a}$

B. $\frac{2b^2 - a^2}{a}$

C. $\frac{2a^2 - b^2}{a^2}$

D. $\frac{a^2 - b^2}{b}$

41. If $\cos \delta = \frac{a}{b}$, find $\tan^2 \delta - 1$.

A. $\frac{b}{a}$

B. $\frac{b^2}{a^2}$

C. $\frac{b^2}{a}$

D. $\frac{a^2}{b}$

42. If $\cos \delta = \frac{a}{b}$, find $\operatorname{cosec}^2 \delta - 1$.

A. $\frac{a^2}{b^2 - a^2}$

B. $\frac{b^2}{b^2 - a^2}$

C. $\frac{b^2}{a - b^2}$

D. $\frac{a^2 - b}{b}$

43. If $\cos \delta = \frac{a}{b}$, find $\operatorname{cosec}^2 \delta + 1$.

A. $\frac{2b^2 - a^2}{b^2 - a^2}$

B. $\frac{b^2}{b^2 - a^2}$

C. $\frac{b^2}{a - b^2}$

D. $\frac{a^2 - b}{b}$

44. If $\cos \delta = \frac{a}{b}$, find $\cot^2 \delta + \cos^2 \delta$.

A. $\frac{2b^2 - a^2}{b^2 - a^2}$

B. $\frac{a^4}{b^2(b^2 - a^2)}$

C. $\frac{b^4}{a - b^2}$

D. $\frac{a^2 - b}{b}$

45. Given that $\tan q = \frac{7}{2}$, evaluate $\sin 2q$.

A. $\frac{28}{35}$

B. $\frac{35}{28}$

C. $\frac{28}{53}$

D. $\frac{53}{28}$

46. Given that $\tan q = \frac{7}{2}$, evaluate $\cos 2q$.

A. $-\frac{53}{45}$

B. $\frac{53}{45}$

C. $\frac{45}{53}$

D. $-\frac{45}{53}$

47. Given that $\tan q = \frac{7}{2}$, evaluate $\tan 2q$.

A. $-\frac{45}{28}$

B. $\frac{54}{29}$

C. $\frac{29}{54}$

D. $-\frac{28}{45}$

48. Given that $\tan q = \frac{7}{2}$, evaluate $\sin 2q - \cos 2q$.

A. $-\frac{53}{73}$

B. $\frac{54}{29}$

C. $\frac{73}{53}$

D. $-\frac{28}{45}$

49. If $\sin \gamma = \frac{a}{1-b}$, find $\cot \gamma$.

- A. $\frac{a}{(1-a-b)(1+a-b)}$
- B. $\frac{(1-a-b)(1+a-b)}{a}+++$
- C. $\frac{b}{(1-a+b)(1+a-b)}$
- D. $\frac{(1-a+b)(1+a-b)}{b}$

50. If $\sin \gamma = \frac{a}{1-b}$, find $\cos \gamma$.

- A. $\frac{(1-a-b)(1+a-b)}{1-b}+++$
- B. $\frac{(1+a-b)(1+a-b)}{1-b}$
- C. $\frac{(1-a-b)(1+a+b)}{1-b}$
- D. $\frac{(1-a+b)(1+a-b)}{1-b}$

SEQUENCES AND SERIES

1. The first term of an arithmetic series is 3, the common difference is 4 and the sum of all term is 820. Find the number of terms and the last term.

- A. 79, 20
- B. 20, 79 +++
- C. -20.5, 80
- D. 80, -20.5

2. Find the sum to which the series converges

$$\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \dots$$

- A. $\frac{1}{2}$
- B. $\frac{1}{4} + \dots$
- C. $\frac{1}{8}$
- D. $\frac{1}{16}$

3. What is the ninth term of the geometric sequence if the third term and the seventh term are -1 and -81?

- A. 3
- B. $-\frac{1}{9}$
- C. -729 +++
- D. 927

4. What is the second term of the series, if the three numbers in geometric sequence whose sum is 13 and whose product is 64

- A. 4 +++

B. 3

C. 5

D. 6

5. Evaluate the tenth term of the series
3+9+27+81+...

- A. 50499
- B. 59490
- C. 59049 +++
- D. 59940

6. Find the three numbers in arithmetic progression whose sum is 3 and whose product is -15.

- A. 5, -3, -1
- B. -3, 1, 5 +++
- C. 1, 5, -3
- D. None

7. If $U_1 = -1$, $U_2 = -5$ and $U_r = a + br$, find a and b .

- A. 2, -3
- B. -1, -5
- C. 3, -4 +++
- D. 4, -3

8. To what sum does the following series converges $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$

- A. $\frac{1}{3}$
- B. 3

C. $\frac{3}{4}, \dots$

D. 4, -3

9. The first term and the last term of a geometric series are 3 and 768, if the sum of the terms is 1533. Find the common ratio.

A. 3

B. $\frac{1}{2}, \dots$

C. 2

D. $-\frac{1}{2}$

10. Find the first term and common ratio of the geometric series if the third term and the seventh term are 81 and 16 respectively.

A. $\frac{2}{3}, \frac{729}{4}, \dots$

B. $\frac{3}{2}, \frac{4}{729}, \dots$

C. $\frac{2}{3}, \frac{4}{729}, \dots$

D. $\frac{3}{2}, \frac{729}{4}, \dots$

11. The sum of the first twenty terms of an arithmetic progression is 45, and the sum of the first forty term is 290. Find the first term and the common difference.

A. $-\frac{5}{2}, 2$

B. $-\frac{2}{5}, \frac{1}{2}$

C. $\frac{1}{2}, -\frac{5}{2}$

D. $-\frac{5}{2}, \frac{1}{2}, \dots$

12. Find the formula for $1, 2, 3, \frac{3}{2}, \frac{3}{4}, \dots$

A. $n!$

B. $16 - 4n$

C. $\frac{24}{2^n}, \dots$

D. $2n + 3$

13. What limit does the sequence

$\left\{ \frac{n^2 + 1}{3n^2 + 4n + 2} \right\}_{n=1}^{\infty}$ converges into?

A. 0

B. ∞

C. $\frac{1}{3}, \dots$

D. 1

14. Find the S_n for the arithmetic sequence $8, 12, 16, 20, \dots$

A. $4n + 4, \dots$

B. $2n(n + 3)$

C. $2n(n - 1)$

D. $2n(n + 4)$

15. Find the sum of all the odd numbers between 1 and 99.

A. 25000

B. 2500, $\dots$

C. 250

D. 250000

16. A model railway manufacturer makes pieces of track of lengths 8cm, 10cm, 12cm etc up to and including 38cm. An enthusiast buys 5 pieces of each length. What is total length of track can be made?

- A. 345+++
- B. 300
- C. 380
- D. 400

17. The general formula for the reciprocal of geometric sequences is...

- A. $\frac{r(r^n-1)}{ar^n(r-1)}$ +++
- B. $\frac{r(r^n-1)}{r^n(r-1)}$
- C. $\frac{(r-1)}{ar(r^n-1)}$
- D. $\frac{d(r^n-1)}{r^n(r-1)}$

18. The sums $1, 1+2, 1+2+3, \dots$ referred to as

- A. Sequence
- B. triangular number +++
- C. amicable number
- D. perfect square

19. Which one of the following is correct for Harmonic progression

- A. $\frac{1}{a+(n-1)d}$ +++
- B. $a+l$
- C. ar^{n-1}

D. $\sqrt{a+(n-1)d}$

20. Find the sum of the first twenty-five odd numbers

- A. 526
- B. 625+++
- C. 265
- D. 562

21. If the second term and fourth term of an exponential function are 6 and 54, find the n^{th} term

- A. $\frac{3^{n-1}}{2}$
- B. 3^{n-1}
- C. $2 \cdot 3^{n-1}$ +++
- D. 6^{n-1}

22. The first term of a geometric progression is 10 and the sum to infinity is 20. Find the common ratio.

- A. 2
- B. $\frac{1}{2}$ +++
- C. $\frac{1}{4}$
- D. 4

23. The first term of an A.P. is 4 while the sum to infinity is 20. Find the common ratio

- A. $\frac{4}{5}$ +++
- B. 4
- C. $\frac{5}{4}$

D. 5

24. If the sixth term of an arithmetic progression is 11 and the first term is 1, find the common difference

A. 2 +++

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. 4

25. The fourth term of an arithmetic progression is 13 while the tenth term is 31. Find the twenty-first term

A. 46

B. 64+++

C. $\frac{1}{3}$

D. 3

26. The second term and the fourth term of a geometric progression are 8 and 32 respectively. What is the sum of the first four terms

A. 2

B. 4

C. 60+++

D. 15

27. Express the recurring decimal $0.\overline{331}$ as a fraction in its lowest term

A. $\frac{33}{99}$

B. $\frac{33\overline{1}}{99\overline{1}}$ +++

C. $\frac{13\overline{1}}{99\overline{1}}$

D. $\frac{13\overline{1}}{99}$

28. For what value of x does the series

$$\frac{1}{1+4x} - \frac{1+x}{(1+4x)^2} + \frac{(1+x)^2}{(1+4x)^3} - \dots$$

A. $\frac{1+x}{(1+4x)^2}$

B. $-\frac{1+x}{(1+4x)}$

C. $-\frac{1}{4} > x > 0, -\frac{2}{5} > x > -\frac{1}{4}$ +++

D. $-\frac{1}{4} > x > 1, -\frac{5}{2} > x > -4$

29. How many terms of the series $2+3+\frac{9}{2}+\dots$ must be taken for the sum to exceed 30?

A. 4

B. 3

C. 6 +++

D. 5

30. Insert three arithmetic mean between 3 and 19. What is the sum of the linear function.

A. 4

B. 15

C. 19

D. 55+++

31. How many terms are there in a sequence of arithmetic progression whose the sum of the first and last term are 4 and 26 is 180 respectively.

- A. $12+++$
- B. 10
- C. 4
- D. 30

32. Find a, b in the sequence $7\frac{1}{2}, a, b, 4$. If the first three term are in arithmetic sequence and the last three are geometric sequence. Find a and b.

- A. $5, \frac{25}{4}$
- B. $\frac{4}{25}, 5$
- C. $5, -\frac{25}{4}$
- D. $\frac{25}{4}, 5_{+++}$

33. Find x, y in the sequence $x, \frac{25}{4}, 5, b$. If the first three terms are in arithmetic sequence and the last three are in geometric sequence. Find x and b.

- A. $4, \frac{25}{4}$
- B. $4, \frac{15}{2}$
- C. $\frac{15}{2}, 4_{+++}$

D. $4, -\frac{15}{2}$

34. An uncle places a sum of money in saving account for a nephew when he was born. On each succeeding birthday, the uncle deposit two times more than on the previous birthday, the total sum of the first 11 deposit is \$20,480. How large was the first deposit?

- A. \$20
- B. \$10 +++
- C. \$100
- D. \$1000

35. A father place a sum of money in a saving account for his daughter when she was born. On each succeeding birthday, the father deposits two more than on the previous birthday without interest. The total sum of the first thirteen deposit is #40,955.00. How large was the last three deposit.

- A. #2591
- B. #1259
- C. #9521+++
- D. #5912

36. Given $U_r = a + (r - 1)d$ in the sequence 2,5,8,11 Find the sum of the first tenth term.

- A. 513
- B. 153
- C. 155+++
- D. 551

37. The sixth and thirteenth terms of an arithmetic progression are 0 and 14 respectively. Find the sum of the first twenty terms.

- A. 0

B. 14

C. -20+++

D. -10

38. Find the geometric mean for consecutive term x, y, z .

A. $y = \sqrt[3]{xz}+++$

B. $y = xz$

C. $z = xy$

D. $y = \frac{z}{x}$

39. Find the arithmetic mean for the consecutive term x, y, z .

A. $y = \frac{x+2}{z}$

B. $y = \frac{x+z}{2}+++$

C. $z = \frac{x+2}{y}$

D. None

40. Find the geometric mean for consecutive term 2,4,8.

A. 2

B. 4+++

C. 8

D. 16

41. Find the arithmetic mean for consecutive term -3,2,7.

A. 2+++

B. 7

C. -3

D. None

42. To what sum does the following series converge $1 - \frac{1}{5} + \frac{1}{25} - \dots$

A. $\frac{1}{5}$

B. $\frac{5}{6}+++$

C. $\frac{6}{5}$

D. $\frac{4}{5}$

43. Find the sum to which the series converges $\frac{1}{6} + \frac{1}{12} + \frac{1}{24} + \dots$

A. $\frac{1}{6}$

B. 2+++

C. 3

D. 4

44. Find the first term and the common difference of the three numbers in an arithmetic sequence whose sum is 12 and whose product is 64.

A. 0,2

B. 4,0 +++

C. 0,4

D. 2,0

45. The third term and the fourth term of an arithmetic sequence are 15 and 5. What is the first term and the common difference?

A. 35,10

B. 10, -35

C. 35, -10 +++

D. -10, -35

46. What is the first term and the common ratio of the geometric sequence whose the third and the seventh term are -1 and -81 ?

A. $-\frac{1}{9}, 3$

B. $3, -\frac{1}{9}$

C. $\frac{1}{9}, -3$

D. $-\frac{1}{9}, -3$

47. What is the first term and the common difference of the arithmetic sequence whose the third and the seventh term are -1 and -81 ?

A. $39, 20$

B. $20, 39$

C. $39, -20$

D. $-20, -39$

48. Express the recurring decimal $0.3131\overline{}$ as a fraction in its lowest term

A. $\frac{31}{99}$

B. $\frac{13}{99}$

C. $\frac{99}{31}$

D. $\frac{31}{91}$

49. Express the recurring decimal $0.2727\overline{}$ as a fraction in its lowest term

A. $\frac{27}{99}$

B. $\frac{72}{99}$

C. $\frac{99}{27}$

D. None

50. Find the first term of the exponential function whose common ratio of the second term and last term are $\frac{1}{3}$ and $\frac{2}{27}$

A. $\frac{2}{3}$

B. $\frac{3}{2}$

C. $-\frac{2}{3}$

D. $-\frac{1}{3}$

SECTION E:
Expansion

Binomial Theorem and

1. What is the coefficient of x^r in the expansion of $(1 + x)^n$?

A. $\binom{n}{r-1}$

B. $\binom{n+1}{r+1}$

C. $\binom{n}{r} +++$

D. $\binom{n-1}{r}$

2. One of the followings has equal value as $\binom{n}{1}$.

A. $\binom{n}{n-1} +++$

B. $\binom{n}{r}$

C. $\binom{n}{r}$

D. $\binom{n-1}{r}$

3. From Pascal's triangle, addition of $\binom{n}{r-1}$ and $\binom{n}{r}$ give

A. $\binom{n+1}{r} +++$

B. $+\binom{n}{r+1}$

C. $+\binom{n+1}{r-1}$

D. $+\binom{n}{r-1}$

4. In notation form, and considering the expansion in ascending powers of x , $(a + x)^n$ can be expressed as

A. $\sum_{r=1}^n \binom{n}{r} x^r a^{n-r}$

B. $\sum_{r=0}^n \binom{n}{r} x^r a^{n-r} +++$

C. $\sum_{r=0}^n \binom{n}{r} x^{n-r} a^{n-r}$

D. $\sum_{r=0}^n \binom{n}{r} x^{n-r} a^r$

5. When we differentiate the expansion of $(1 + x)^n$ with respect to x two times, put $x = 0$ and simplify, we get

A. $\binom{n}{r} = \frac{2}{n}$

B. $\binom{n}{1} = \frac{2}{n+1}$

C. $\binom{n}{2} = \frac{n(n-1)}{2} +++$

D. $\binom{n}{2} = \frac{2}{n(n-1)}$

6. For positive integer n , the factorial of n , denoted by $n!$ is equivalent to

A. $(n-1)!$

B. $n(n+1)(n-2)!$

C. $(n+1)!$

D. $n(n-1)(n-2)! +++$

7. The value of $\sum_{k=0}^n \binom{n}{k}$ is

A. 2^{n-2}

B. $2^n +++$

C. $2n$

D. n^2

8. Expand $(x + 2y)^7$ in ascending powers of x

A. $x^7 + 14x^6y + 84x^5y^2 + 280x^4y^3 + 560x^3y^4 + 672x^2y^5 + 448xy^6 + 128y^7$

B. $128y^7 + 448xy^6 + 672x^2y^5 + 560x^3y^4 + 280x^4y^3 + 84x^5y^2 + 14x^6y + x^7 +++$

C. $x^7 + 4x^6y + 84x^5y^2 + 28x^4y^3 + 56x^3y^4 + 672x^2y^5 + 448xy^6 + 128y^7$

D. $128y^7 + 448xy^6 + 62x^2y^5 + 560x^3y^4 + 280x^4y^3 + 8x^5y^2 + 14x^6y + x^7$

9. Find the term independent of y in the expansion of $\left(\frac{x^4}{y^3} + \frac{y^2}{2x}\right)^{10}$

A. $\frac{105}{32}x^{11}$

B. $\frac{15}{32}x^{10}$

C. $\frac{105}{32}x^{10} + \dots$

D. $105x^{10}$

10. Find the maximum coefficient in the expansion of $(3x + 5)^{10}$

A. 26581250

B. 26578250

C. 26578125

D. 265781250 + ...

11. Obtain the first four terms of the expansion of $(1 + 3x)^{\frac{1}{3}}$ in ascending powers of x

A. $1 + x - x^2 + \frac{5}{3}x^3 + \dots$

B. $1 - x - x^2 + \frac{5}{3}x^3$

C. $1 - x + x^2 + \frac{5}{3}x^3$

D. $1 + x - x^2 - \frac{3}{5}x^3$

12. Obtain the expansion of $\sqrt{9 + x^2}$ as far as the term in x^4

A. $3 + \frac{1}{3}x^2 + x - \frac{1}{216}x^4$

B. $3 + \frac{1}{6}x^2 - \frac{1}{216}x^4 + \dots$

C. $3 - \frac{1}{6}x^2 - \frac{1}{216}x^4$

D. $3 + \frac{1}{6}x^2 + \frac{1}{216}x^4$

13. Find the constant term in the expansion of $\left(2z^2 + \frac{1}{z}\right)^9$

14. Find the middle term in the expansion of $(x^2 + y^2)^8$.

15. Find the fifth term in the expansion of $(-3x - 4y)^6$ in descending order of x .

16. For what value of x is the expansion of $\frac{1}{a+bx}$ valid?

A. $|x| < \frac{a}{b} + \dots$

B. $|x| > \frac{a}{b}$

C. $|x| < \frac{1}{b}$

D. $|x| < \frac{a}{2}$

17. Find the sum of the constant coefficients in the expansion of $(3 + 2x)^4$

- A. 620
- B. 630
- C. 625 +++
- D. 216

18. Find the first four terms in the expansion of $(1 - 3x^2)^5$ in ascending powers of x

- A. $1 - 15x^2 + 90x^4 - 270x^6$ +++
- B. $1 + 15x^2 + 90x^4 - 270x^6$
- C. $1 - 15x^2 - 90x^4 + 270x^6$
- D. $1 + 15x^2 + 90x^4 + 270x^6$

19. State the value of $|x|$ for which the expression $(27 - 6x)^{\frac{-2}{3}}$ can be expanded as a series of ascending powers of x .

- A. $|x| < \frac{2}{9}$
- B. $|x| < \frac{9}{2}$ +++
- C. $|x| < \frac{1}{9}$
- D. $|x| > \frac{2}{9}$

20. State the condition under which the expansion of $(x + 2y)^{-5}$ will be valid in ascending powers of y .

- A. $\left|\frac{y}{x}\right| < \frac{1}{2}$ +++
- B. $\left|\frac{y}{x}\right| < 1$
- C. $\left|\frac{y}{x}\right| > \frac{1}{2}$
- D. $\left|\frac{x}{y}\right| < \frac{1}{2}$

21. Simplify $\frac{\binom{n}{r+1}}{\binom{n}{r}}$

- A. $\frac{r+1}{n-r}$
- B. $n - r$
- C. $r + 1$
- D. $\frac{n-r}{r+1}$ +++

22. Find the term independent of y in the expansion of $\left(\frac{4}{x^5} - \frac{2x}{3y}\right)^{10}$

- A. $\frac{1048576}{x^{50}}$ +++
- B. $\frac{1048576}{x^{20}}$
- C. $\frac{1048576}{y^{30}}$
- D. $\frac{1048576}{x^{30}}$

23. Find the fifth term in the expansion of $(3x + 2y^2)^{12}$ in descending powers of x

- A. $5196312x^8y^8$
- B. $51963120x^6y^9$
- C. $51963120x^7y^8$
- D. $51963120x^8y^8$ +++

24. Find the value of $\frac{\binom{10}{r}}{\binom{10}{r+1}}$

- A. $\frac{r+1}{10-r}$ +++
- B. $\frac{r}{10-r}$
- C. $\frac{r-1}{10+r}$
- D. $\frac{r}{10+r}$

25. Evaluate the term independent of x in the binomial expansion of $\left(x^2 - \frac{1}{2x}\right)^9$

- A. $\frac{16}{21}$
- B. $\frac{21}{16} +++$
- C. 21
- D. 16

26. Find the constant term in the expansion of $\left(x^2 - \frac{2}{x}\right)^6$

- A. 360
- B. 240 +++
- C. 140
- D. 420

27. Find the value of $\sum_{k=1}^5 k \binom{5}{k}$

- A. 60
- B. 6
- C. 32
- D. 80 +++

28. Find the value of $\sum_{r=3}^8 r! \binom{8}{r}$

- A. 10802
- B. 10956
- C. 109536 +++
- D. 10804

29. Find the coefficient of x^4 in the expansion of $\left(2x - \frac{3}{x^2}\right)^4$

- A. 16 +++

B. 32

C. 42.

D. 14

30. Determine the greatest coefficient in the expansion of $(3x + 1)^8$

- A. 17496
- B. 20412 +++
- C. 2941
- D. 204120

31. Find the constant term in the expansion of $(3x + 1)^8$

- A. 1 +++
- B. 2
- C. 3
- D. 4

32. Find the value of x for which the expansion of $\left(2 + \frac{1}{4}x\right)^5$ is valid

- A. $|x| < 2 +++$
- B. $|x| < 4$
- C. $|x| < 8$
- D. $|x| < \frac{1}{2}$

33. Evaluate $\sum_{r=1}^5 (r + 3)!$

- A. 46224 +++
- B. 4624
- C. 4224
- D. 46228

34. Simplify $\binom{n+1}{n-1} + \binom{n}{n-1}$

- A. $\frac{2}{n+3}$
- B. $\frac{2}{n^2+3}$
- C. $\frac{n^2+3n}{2} +++$
- D. $\frac{n^2-3n}{2}$

35. Find the constant term in the expansion of $\left(\frac{1}{\beta^2} - \beta\right)^{18}$

- A. 18564 +++
- B. 1856
- C. 64531
- D. 185640

36. Simplify $\binom{n+1}{n-1} - \binom{n}{n-1}$

- A. $\frac{n^2-3n}{2}$
- B. $\frac{n^2+3n}{2}$
- C. $\frac{n(n+1)}{2}$
- D. $\frac{n(n-1)}{2} +++$

37. Simplify $\frac{\binom{n+1}{n-1}}{\binom{n}{n-1}}$

- A. $\frac{n-1}{2}$
- B. $\frac{n+1}{2}$
- C. $\frac{n}{2}$
- D. $\frac{n}{2} + 1 +++$

38. Find the term independent of y in the expansion of $\left(\frac{x^4}{3y^3} + \frac{y^2}{2x}\right)^5$

- A. $\frac{5}{36}x^5 +++$
- B. $\frac{5}{6}x^5$
- C. $\frac{36}{5}x^5$
- D. $36x^5$

39. Find the coefficient of x^{14} in the expansion of $(20x^2 - 1)^8$

- A. 10240000000
- B. 10240000
- C. $-10240000000 +++$
- D. 1024000000

40. Take away the coefficient of x^{14} in the expansion of $(20x^2 - 1)^8$ from the coefficient of x^{14} in the expansion of $(1 + x)^{14}$

- A. 10240000001 +++
- B. 1024000
- C. 10240000
- D. 1024000001

41. Express n in factorial form.

- A. $n!$
- B. $(n-1)!$
- C. $\frac{n!}{(n-1)!} +++$
- D. $\frac{n!}{(n-1)}$

42. Find the binomial expansion of $\left(1 + \frac{3}{4}x\right)^4$ in descending powers of x .

A. $\frac{81}{256}x^4 + \frac{27}{16}x^3 + \frac{27}{8}x^2 + 3x + 1 + \dots$

B. $1 + 3x + \frac{27}{8}x^2 + \frac{27}{16}x^3 + \frac{81}{256}x^4$

C. $\frac{81}{256}x^4 - \frac{27}{16}x^3 + \frac{27}{8}x^2 - 3x + 1$

D. $\frac{81}{256}x^4 - \frac{27}{16}x^3 - \frac{27}{8}x^2 + 3x - 1$

43. Find the coefficient of x^4 in the expansion of $\left(1 + \frac{3}{4}x\right)^4$

A. $\frac{8}{25}$

B. $\frac{81}{25}$

C. $\frac{81}{256} + \dots$

D. $\frac{81}{26}$

44. Find the coefficient of x in the expansion of $\left(1 + \frac{3}{4}x\right)^4$

A. 1

B. 2

C. 3 + ...

D. 31

45. Find the value of n when $\binom{n}{2} = 120$.

A. -15

B. 15

C. -16

D. 16 + ...

46. Express $\binom{n+r}{n-r}$ in factorial form

A. $\frac{(n+r)!}{(2r)!(n-r)!} + \dots$

B. $\frac{(n+r)!}{(2r)!(n-r+1)!}$

C. $\frac{n!}{2!r!}$

D. $\frac{(n-r)!}{(2r)!(n+r)!}$

47. Find the value of n when $\binom{n-2}{1} = 5$

A. 5

B. 6

C. 7 + ...

D. 8

48. Express $\frac{n(n-1)}{2}$ in factorial form

A. $\frac{n(n-1)!}{2!}$

B. $\frac{n(n+1)!}{2!}$

C. $\frac{n!}{2!}$

D. $\frac{n!}{(n-2)!2!} + \dots$

49. Find the term independent of y in the expansion of $\left(\frac{1}{2}x + 3y\right)^{1000}$

A. $\left(\frac{x}{2}\right)^{1000} + \dots$

B. $\frac{x}{2^{1000}}$

C. $\left(\frac{x}{2}\right)^{100}$

D. $\frac{x}{2^{100}}$

50. In the expansion of $\left(\frac{1}{2}x + 3y\right)^{1000}$, find the coefficient of x in the term independent of y .

A. $\frac{1}{2^{100}}$

B. $\frac{1}{2^{1000}} + \dots$

C. $\frac{x}{2^{100}}$

D. $\frac{x}{2^{1000}}$